

Arpit Raj

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EDUCATION

- **LNM Institute of Information Technology, India** August 2024 – April 2028
Bachelor of Technology in Electronics and Communication Engineering CGPA: 7.67
Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Programming for Problem Solving

SKILLS SUMMARY

- **Languages:** C, C++, Python, JavaScript, SQL, VHDL, Bash, Go, Kotlin
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js, PyTorch, TensorFlow, Keras, NumPy, Pandas, Hugging Face Transformers, Scikit-learn, Matplotlib, OpenCV, NLTK, spaCy, LangChain, LangGraph, FastAPI, Flask, Django
- **Tools:** Git, GitHub, Docker, Kubernetes, MySQL, MongoDB, Postman, Jenkins, VS Code, Jupyter Notebook, Redis, anaconda, Power BI, Tableau, Figma, Vivado, Arduino IDE, MATLAB
- **Platforms:** Linux, Windows, AWS, Netlify, Vercel, ESP32

EXPERIENCE

- **LNM Institute of Information Technology** Jaipur, India
Teaching Assistant – Programming for Problem Solving (PPS) Aug 2025 – Jan 2026

Technologies: C, C++, Data Structures, Algorithms, Debugging

- * **Teaching & Mentorship:** Mentored over 240+ undergraduate students through tutorials, laboratory sessions, and doubt-solving sessions as part of the institute's Teaching Assistantship program.
- * **Programming Instruction:** Conducted 30+ hours of academic guidance covering C/C++, programming fundamentals, problem-solving techniques, recursion, pointers, dynamic memory allocation, and data structures.
- * **Code Review & Debugging:** Guided students in writing clean, modular, and efficient code by reviewing implementations, identifying logical errors, and teaching debugging and optimization techniques. Assisted in evaluating 10+ laboratory assignments and examinations.

- **Mentox Technologies Pvt. Ltd.** Jaipur, India
Summer Intern – Full Stack Developer May 2026 – Jul 2026

Tech Stack: React, Node.js, Express.js, MySQL, HTML, CSS, JavaScript, REST APIs, Git, OAuth, JWT, multer

- * **Workshop Management Portal:** Designed and developed a full-stack web application to digitalize the company's workshop and bootcamp management process, providing a centralized platform for administrators and participants.
- * **Backend Development:** Designed a scalable backend with 39 REST endpoints across 8 controllers, implementing role-based Google OAuth 2.0 authentication, JWT session management, file uploads using Multer, assignment workflows, and 12 real-time Socket.io events for notifications and live updates.
- * **Frontend Development:** Built 2 React applications with 13,800+ LOC, 28 reusable components, including student dashboard with live leaderboards, assignment management, feed management, admin controls, doubts discussion forum, Framer Motion and GSAP animations, and an analytics-driven admin portal.
- * **System Design & Database:** Designed a relational schema with 12 normalized Sequelize models supporting users, assignments, doubts, discussions, notifications, polls, comments, and grading workflows, enabling end-to-end digital management of workshops and bootcamps.

PROJECTS

- **Noise-Robust Multimodal Video Retrieval** : Created a multimodal retrieval system that converts videos into semantically searchable timestamps, enabling natural language queries even under severe transmission noise or low-quality cameras. Integrates semantic communication principles with VLMs.
Tech Stack: Python, PyTorch, Transformers, CLIP, BLIP-2, OpenCV, FAISS, Streamlit, NumPy, scikit-learn.
- **MedSync : AI-powered Rural Healthcare Platform:** Built a full-stack healthcare platform supporting doctor-specialization retrieval using RAG, appointment scheduling, WebRTC-based video consultations, transcript-driven prescription digitization, automated medicine synchronization based on transcript, and a Retrieval-Augmented Generation (RAG) chatbot for pre-consultation assistance and medical triage, featuring 20+ REST APIs and 8+ relational tables
Tech Stack: React.js, Node.js, Express.js, MySQL, WebRTC, Python, LangChain, RAG, REST APIs, JavaScript, CSS.
- **Autonomous Driving Agent using CARLA:** Developed an end-to-end autonomous driving pipeline using a custom PyTorch U-Net trained on 30,000+ images based on CARLA simulator data for real-time semantic lane segmentation and achieved 92% IoU. Extracted lane boundaries from the segmentation masks and utilized a PID controller to compute steering commands at 20 FPS for autonomous lane following.
Tech Stack: Python, PyTorch, CARLA Simulator, OpenCV, NumPy, PID Controller.

PUBLICATIONS

- **Deep Generative Semantic Communication: Channel-Aware Language Models** : Authored a research paper proposing a channel-aware Transformer-based semantic communication framework for 6G networks. Introduces SNR-conditioned adaptation for robust semantic transmission over AWGN channels and analyzes representation geometry using Effective Rank, CKA, and Information Bottleneck metrics.
Tech Stack: Python, PyTorch, Hugging Face Transformers, NLP, Deep Learning, Semantic Communication